

# ONLINE FOOD ORDERING SIMULATION

## Abstract

The Online Food Ordering Simulation is a console-based application developed in the C programming language that simulates the process of ordering food in an online environment. The main objective of this project is to provide a simple digital platform where users can view a menu, select food items, specify quantity, place multiple orders, and receive the final bill automatically.

In traditional ordering systems, customers manually place orders and billing is often calculated separately, which can consume time and may lead to calculation errors. This application automates the ordering and billing process by providing a structured menu-driven system. The user can choose food items such as burgers, pizza, sandwiches, biryani, cool drinks, and ice cream, enter the required quantity, and the program calculates the bill instantly.

The project demonstrates important programming concepts such as variables, loops, conditional statements, switch-case, user input/output operations, and arithmetic calculations. It is designed to be simple, user-friendly, and easy to understand, making it suitable for academic mini-projects and beginners learning C programming.

Overall, the Online Food Ordering Simulation is a practical application that showcases how programming can simplify restaurant ordering systems and billing operations.

## Introduction

With the rapid growth of technology and internet services, online food ordering has become a common part of modern life. Customers now prefer ordering food digitally rather than visiting restaurants physically. Online ordering systems provide convenience, save time, and make billing more efficient.

The Online Food Ordering Simulation project is developed to imitate a basic online food ordering platform using the C programming language. It is a menu-driven application where users can select items from a predefined food menu, enter quantity, and generate a bill based on their choices.

The system allows customers to:

- View food menu with prices
- Select desired food items

Enter quantity required

Order multiple items in one session

View running total bill

Get final billing amount after ordering

This project mainly focuses on understanding programming logic and applying it to a real-world application. It uses:

Loops for repeated ordering

Switch-case statements for menu selection

Variables for billing calculation

Input/output functions for user interaction

The project can act as a foundation for developing advanced restaurant management or food delivery applications in the future.

## **Algorithm**

Algorithm for Online Food Ordering Simulation

Step 1: Start the program.

Step 2: Declare variables:

choice

quantity

total bill

current bill

continue option

Step 3: Display food menu with prices.

Step 4: Ask user to enter food choice.

Step 5: Ask user to enter quantity.

Step 6: Check selected choice using switch-case.

Step 7: Calculate bill based on: Price  $\times$  Quantity

Step 8: Add current bill to total bill.

Step 9: Display current bill amount.

Step 10: Ask user whether to order more.

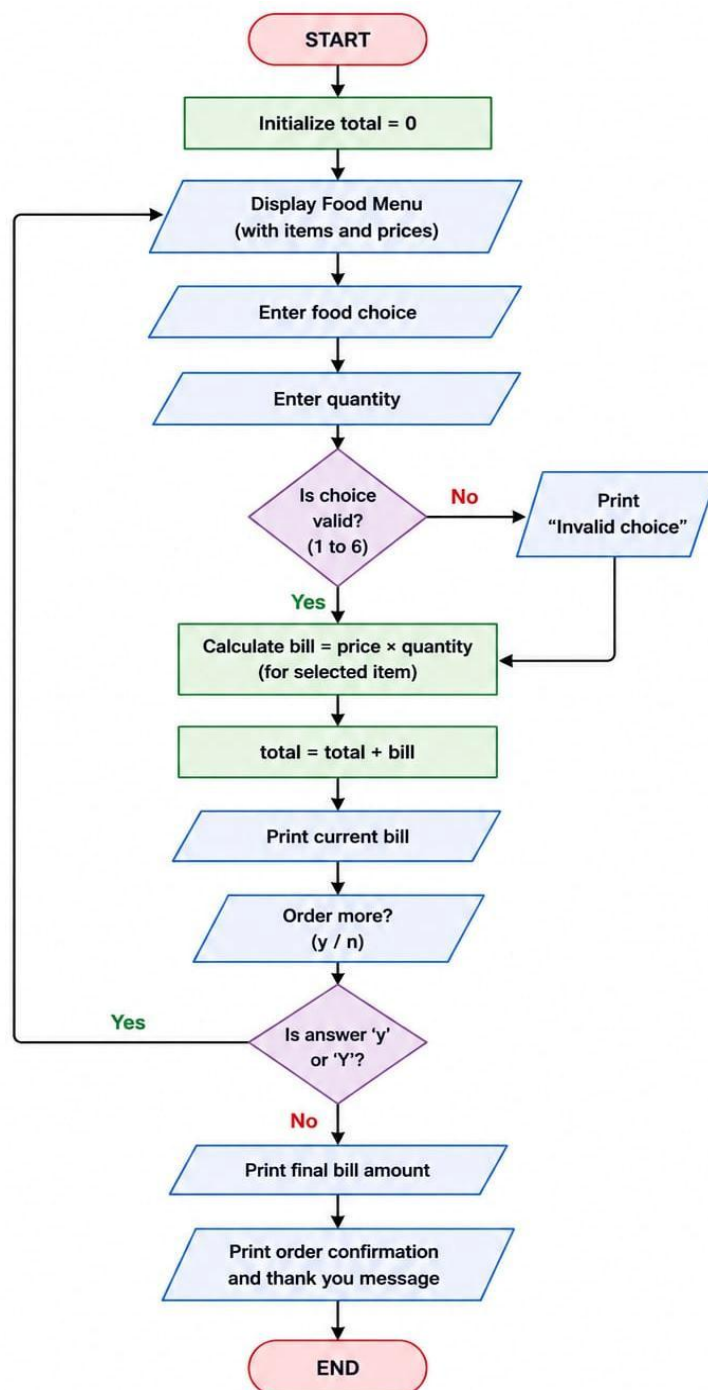
Step 11: If user enters 'y': Repeat from Step 3.

Step 12: Otherwise: Display final bill amount.

Step 13: Display order confirmation message.

Step 14: Stop the program.

## Flowchart



## **Methodology**

The development of Online Food Ordering Simulation follows a systematic methodology.

### **Requirement Analysis -**

The system requirements include:

- Food menu display
- Customer selection
- Quantity input
- Bill generation
- Multiple ordering option
- Final bill calculation

### **System Design -**

The application is designed as a menu-driven console program.

Main components:

- Menu Display Module
- Order Selection Module
- Billing Module
- Confirmation Module

### **Implementation -**

The program is implemented using C programming concepts:

### **Input Module -**

Takes:

- Food choice
- Quantity
- Continue option

### Processing Module -

Calculates:

- Item cost

- Running total bill

### Decision Module -

Uses switch-case for:

- Valid item selection

- Invalid choice handling

### Looping Module -

- Allows repeated ordering using do-while loop.

### Output Module -

Displays:

- Added item message

- Current bill

- Final bill

- Thank you message

### Testing -

The system is tested for:

- Valid choice

- Invalid choice

- Multiple orders

- Bill accuracy

- Exit condition

## Results

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^ ONLINE FOOD ORDERING SIMULATION

Menu:
1. Burger - Rs.120
2. Pizza - Rs.250
3. Sandwich - Rs.80
4. Biryani - Rs.180
5. Cool Drink - Rs.50
6. Ice Cream - Rs.90
Enter your choice: 5
Enter quantity: 2
Cool Drink added successfully
Current Bill = Rs. 100.00
Do you want to order more? (y/n): n
Final Bill Amount = Rs. 100.00
Order Placed Successfully
Thank You for Ordering

=== Code Execution Successful ===
```

```
Output

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Menu:
1. Burger - Rs.120
2. Pizza - Rs.250
3. Sandwich - Rs.80
4. Biryani - Rs.180
5. Cool Drink - Rs.50
6. Ice Cream - Rs.90
Enter your choice: 6
Enter quantity: 3
Ice Cream added successfully
Current Bill = Rs. 270.00
Do you want to order more? (y/n): n
Final Bill Amount = Rs. 270.00
Order Placed Successfully
Thank You for Ordering

=== Code Execution Successful ===
```

```
Output

^ ONLINE FOOD ORDERING SIMULATION

Menu:
1. Burger - Rs.120
2. Pizza - Rs.250
3. Sandwich - Rs.80
4. Biryani - Rs.180
5. Cool Drink - Rs.50
6. Ice Cream - Rs.90
Enter your choice: 7
Enter quantity: 5
Invalid choice
Current Bill = Rs. 0.00
Do you want to order more? (y/n): n
Final Bill Amount = Rs. 0.00
Order Placed Successfully
Thank You for Ordering

=== Code Execution Successful ===
```

## Conclusion

The Online Food Ordering Simulation project successfully demonstrates a basic restaurant ordering and billing system using C programming. It provides a simple and efficient way for customers to select food items, specify quantity, and generate a final bill automatically.

The project eliminates manual billing calculations, improves speed, and ensures accurate pricing. It also demonstrates practical implementation of programming concepts such as loops, conditional statements, switch-case, and arithmetic operations.

This project is useful as an educational tool for understanding how programming logic can be applied to real-life applications. It is simple, user-friendly, and serves as a strong foundation for more advanced food ordering systems.

Overall, the project proves that digital ordering systems can improve convenience, efficiency, and customer experience in the food service industry.

## Future work

The system can be improved further by adding advanced features such as:

### Customer Registration -

Allow customer login/signup.

### Cart System -

Add multiple items to cart before checkout.

### Discount Offers -

Apply:

Coupons

Festival discounts

Combo offers

### Payment Gateway -

Support:

UPI

Credit Card

Debit Card

Cash on Delivery

### Order Tracking -

Show:

Order confirmed

Preparing

Out for delivery

Delivered

### Restaurant Selection -

Allow ordering from multiple restaurants.

### Database Storage -

Store:

Customer details

Order history

Billing records

### Graphical User Interface -

Develop using:

C++

Java

Python

Web technologies

### Mobile App Version -

Create Android/iOS app.

### AI Recommendation -

Suggest food based on:

Previous orders

Customer preferences

Popular dishes



